

1.8 Vector Norm



Core skills you will master in this section:

- ▶ Vector norm: provides a unified standard that maps a high-dimensional vector to a scalar that can be compared in magnitude.
- ▶ Three basic norms: L^1 (sum of absolute values), L^2 (Pythagorean theorem), L^∞ (largest absolute value among the components).
- ▶ L^p norm form: a continuous generalization from L^1 to L^∞ .
- ▶ Geometric understanding of norms: use planar contour plots to visually understand the shape characteristics and trends of different norms.
- ▶ Vector norms and distance: different norms correspond to different distance metrics, such as Euclidean distance and Manhattan distance.
- ▶ Pairwise distance matrix structure: using a matrix to represent the distances between pairs of points.

In high-dimensional space, the magnitude of a vector cannot be compared directly, so vector norms are introduced to measure their size. Common vector norms include the L^1 , L^2 , and L^∞ norms. The L^1 norm is the sum of the absolute values of the vector's components, the L^2 norm is the Euclidean distance, and the L^∞ norm is the largest absolute value among the vector's components. Building on these, this section goes on to introduce the L^p norm, and reinforces the geometric perspective on these vector norms.

Which Vector Is Larger? Which Is Smaller?

Simply put, vector norms make it possible to compare the size of vectors.

On a number line, it's easy to see that 5 is larger than 4; even when the signs differ — say, comparing -5 and 4 — we can take the absolute value (absolute value) and then compare. The absolute value is, in effect, the distance between the number and the origin.

However, in two-dimensional and higher-dimensional spaces, the magnitude of a vector cannot be compared directly. For example, given vectors $\mathbf{a} = \begin{bmatrix} 3 \\ -4 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$, which one is larger? Which is smaller?

To make comparison possible, we introduce the vector norm; simply put, a vector norm provides a way to measure the size of a vector — it maps a vector to a scalar according to a specific rule, and we then compare those scalars.

Let's first look at a few common vector norms.

L^1 Norm

The L^1 norm, also known as the city block distance or Manhattan distance, computes the sum of the absolute values of the vector's components, that is,

$$\|\mathbf{x}\|_1 = \sum_{i=1}^n |x_i|$$

For the 2D vector $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, the L^1 norm is:

$$\left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right\|_1 = |x_1| + |x_2|$$

For example, given the vector $\mathbf{a} = [3, -4]^T$, its L^1 norm is computed as follows:

$$\|\mathbf{a}\|_1 = |3| + |-4| = 3 + 4 = 7$$

As another example, given the vector $\mathbf{b} = [0, 5]^T$, its L^1 norm is computed as follows:

$$\|\mathbf{b}\|_1 = |0| + |5| = 0 + 5 = 5$$

By the L^1 norm's calculation rule, vector $\mathbf{a}$ is larger.

Code 1 computes the L1 norm. Please pay attention to the following two lines.

a `numpy.linalg.norm()` computes the vector norm; `ord=1` means computing the L^1 norm, i.e., summing the absolute values of all elements of the vector.

b This verifies the result of the previous statement. `numpy.abs(a)` computes the absolute value of each element in vector `a`; `numpy.sum([3, 4])` computes the sum of all elements in the array, giving a result of 7.

Code 1. L^1 Norm |  LA_Ch08_01_01.ipynb

```
## 初始化
import numpy as np

## 定义向量
a = np.array([3, -4])
b = np.array([0, 5])

## 计算 L1 范数
a L1_a = np.linalg.norm(a, ord=1)
b np.sum(np.abs(a))

L1_b = np.linalg.norm(b, ord=1)
np.sum(np.abs(b))
```

As shown in Figure 1, for the 2D vector $\mathbf{x} = [x_1, x_2]^T$, the L^1 -norm iso-distance line (iso-distance line) in the plane is a rotated square (a special rhombus), satisfying

$$|x_1| + |x_2| = c$$

Here, c is a given L^1 norm value, with $c \geq 0$.

Figure 1's black arrows represent $\mathbf{e}_1 = [1, 0]^T$ and $\mathbf{e}_2 = [0, 1]^T$. Both $\mathbf{e}_1$ and $\mathbf{e}_2$ have an L^1 norm of 1. In the first quadrant ($x_1 > 0, x_2 > 0$), the 2D vector $\mathbf{x} = [x_1, x_2]^T$ has an L^1 norm of 1 precisely when $x_1 + x_2 = 1$.

 Please try to write out the analytic expression for in different quadrants.

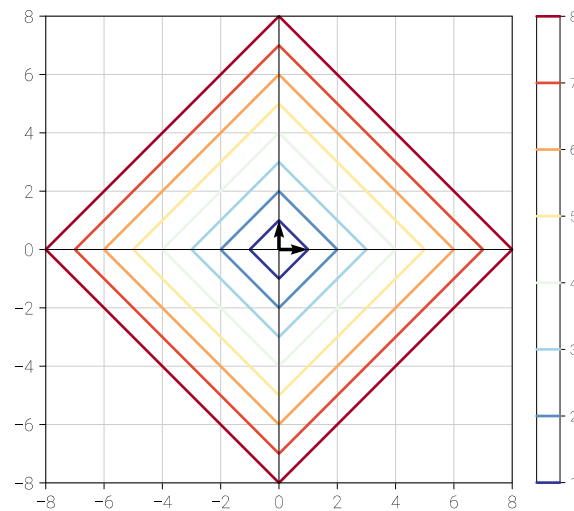


Figure 1. Iso-Distance Lines of the L^1 Norm for a 2D Vector in the Plane

Code 2 plots Figure 1; let's walk through its key lines below.

Code 2. Plotting the Iso-Distance Lines of the L^1 Norm in the Plane |  LA_01_08_02.ipynb

```

## 初始化
import numpy as np
import matplotlib.pyplot as plt

## 生成数据
a xx, yy = np.meshgrid(np.linspace(-8, 8, 1000),
                        np.linspace(-8, 8, 1000))

## 计算 L1 范数
b zz = np.abs(xx) + np.abs(yy) # L1 范数计算公式

## 定义标准基向量
e1 = np.array([1, 0])
e2 = np.array([0, 1])

## 可视化
plt.figure(figsize=(6, 6))

# 绘制标准基向量 e1 和 e2
c plt.quiver(0, 0, e1[0], e1[1],
             angles='xy', scale_units='xy',
             scale=1, color=[0, 0, 0],
             label='e1', zorder=1000)

plt.quiver(0, 0, e2[0], e2[1],
             angles='xy', scale_units='xy',
             scale=1, color=[0, 0, 0],
             label='e2', zorder=1000)

# 绘制 L1 范数的等距线
d contour = plt.contour(xx, yy, zz,
                        levels=np.arange(1, 9),
                        cmap='RdYlBu_r')

# 添加 colorbar
cbar = plt.colorbar(contour)

# 设置坐标轴范围
plt.xlim(-8, 8); plt.ylim(-8, 8)

# 绘制网格和坐标轴
plt.axhline(0, color='black', linewidth=0.5)
plt.axvline(0, color='black', linewidth=0.5)
plt.xticks(np.arange(-8, 9, 2)); plt.yticks(np.arange(-8, 9, 2))
plt.grid(True, which='both', linestyle='--',
         linewidth=0.5, color='0.8')

# 设置等比例显示
plt.gca().set_aspect('equal', adjustable='box')

```



a `numpy.meshgrid()` generates 2D grid arrays, which together make up all the coordinate points in the plane. This process is like tiling the plane with a grid, preparing to perform a calculation at every point.

b Computes the L1 norm at each grid point. `np.abs(xx)` takes the absolute value of every number in `xx`, and `np.abs(yy)` does the same for `yy`. Adding them together gives `zz`, representing the L1 norm value at each point. The L1 norm can simply be understood as “adding together the absolute values of the horizontal and vertical coordinates.”

c Plots the vectors `e1` and `e2`.
`plt.quiver()` is the function used to draw arrows.
 The first two parameters `(0, 0)` are the arrow's starting point — that is, it starts from the origin.
`e1[0]` and `e1[1]` give the arrow's direction `(1, 0)`, meaning it points in the positive x-axis direction.
`angles='xy'` and `scale_units='xy'` tell the program that the arrow's direction and units are based on the coordinate axes.
`scale=1` means no scaling is applied.
`color=color_e1` is black, `label='e1'` adds a label to the arrow, and `zorder=1000` places this arrow on top so it isn't hidden behind other plot elements.

d Plots the contour lines (iso-distance lines) of the L1 norm.
`plt.contour()` is the function used to draw contour lines — much like drawing “contour lines” on a map, it connects points that share the same norm value.
`xx` and `yy` are the horizontal and vertical coordinate grids, and `zz` is the value at each point, namely the L1 norm.
`levels=np.arange(1, 9)` means drawing a contour line for each level from 1 to 8.
`cmap='RdYlBu_r'` means using the “red-yellow-blue, reversed” colormap to fill the different levels, where `_r` indicates the color order is reversed.

L² Norm

The L^2 norm, also known as the Euclidean norm or the modulus, computes the length (magnitude) of a vector:

$$\|\mathbf{x}\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$$

This is the most commonly used measure in the physical world — for example, measuring the straight-line distance between points in space.

For the 2D vector $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, the L^2 norm is:

$$\left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right\|_2 = \sqrt{x_1^2 + x_2^2}$$

For example, given the vector $\mathbf{a} = [3, -4]^T$, its L^2 norm is:

$$\|\mathbf{a}\|_2 = \sqrt{3^2 + (-4)^2} = \sqrt{9+16} = \sqrt{25} = 5$$

As another example, given the vector $\mathbf{b} = [0, 5]^T$, its L^2 norm is computed as follows:

$$\|\mathbf{b}\|_2 = \sqrt{0^2 + 5^2} = \sqrt{0+25} = \sqrt{25} = 5$$

By the L^2 vector norm's calculation rule, vectors $\mathbf{a}$ and $\mathbf{b}$ are equal.

As shown in Figure 2, for the 2D vector $\mathbf{x} = [x_1, x_2]^T$, the L^2 -norm iso-distance line in the plane is a perfect circle, satisfying

$$\sqrt{x_1^2 + x_2^2} = c$$

Here, c is a given L^2 norm value, with $c \geq 0$.

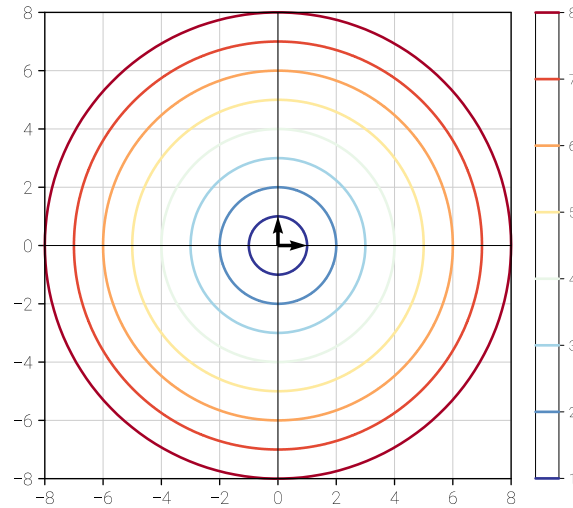


Figure 2. Iso-Distance Lines of the L^2 Norm for a 2D Vector in the Plane



LA_01_08_03.ipynb computes and compares vectors $\mathbf{a}$ and $\mathbf{b}$'s L^2 norms; LA_01_08_04.ipynb plots Figure 2. Both code files are simple, so please study them on your own.

L^∞ Norm

The L^∞ norm, also known as the Chebyshev norm, computes the largest absolute value among the vector's components, that is,

$$\|\mathbf{x}\|_\infty = \max_i |x_i|$$

For the 2D vector $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, the L^∞ norm is:

$$\left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right\|_\infty = \max(|x_1|, |x_2|)$$

For example, given the vector $\mathbf{a} = [3, -4]^T$, its L^∞ norm is computed as follows:

$$\|\mathbf{a}\|_\infty = \max(|3|, |-4|) = 4$$

As another example, given the vector $\mathbf{b} = [0, 5]^T$, its L^∞ norm is computed as follows:

$$\|\mathbf{b}\|_\infty = \max(|0|, |5|) = 5$$

By the L^∞ vector norm's calculation rule, vector $\mathbf{b}$ is larger.

As shown in Figure 3, for the 2D vector $\mathbf{x} = [x_1, x_2]^T$, the L^∞ -norm iso-distance line in the plane is a square, satisfying

$$\max(|x_1|, |x_2|) = c$$

Here, c is a given L^∞ norm value, with $c \geq 0$.

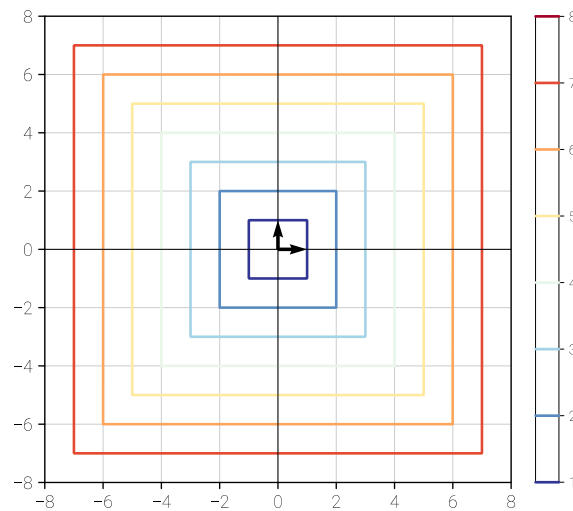


Figure 3. Iso-Distance Lines of the L^∞ Norm for a 2D Vector in the Plane

? Please try writing code to plot Figure 3.



LA_01_08_03.ipynb computes and compares vectors $\mathbf{a}$ and $\mathbf{b}$'s L^∞ norms; note that positive infinity is written as `numpy.inf`.

L^p Norm

To unify these different norms, we can define the L^p norm:

$$\|\mathbf{x}\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{\frac{1}{p}}$$

Note that $p \geq 1$.

For the 2D vector $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$, the L^p norm is:

$$\left\| \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right\|_p = \left(|x_1|^p + |x_2|^p \right)^{\frac{1}{p}}$$

As shown in Figure 4, in the plane, as p takes on different values, $\|\mathbf{x}\|_p = 1$ corresponds to different shapes.

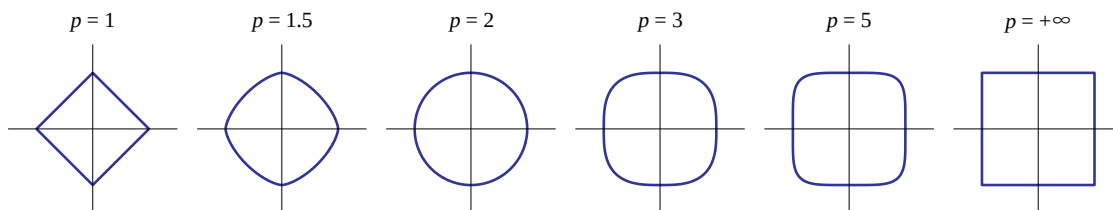


Figure 4. As p Takes on Different Values in the Plane, the L^p Norm Corresponds to Different Shapes

Figure 5 shows vectors $\mathbf{a} = [3, -4]^T$ and $\mathbf{b} = [0, 5]^T$, and how their L^p norms change as p varies.

As p increases, for the vector $\mathbf{a} = [3, -4]^T$, its L^p norm keeps decreasing. When p is close to 1, the contributions of all the vector's components are taken into account; as p tends to $+\infty$, the component with the largest absolute value alone determines the norm value.

As for the vector $\mathbf{b} = [0, 5]^T$, its L^p norm doesn't change as p varies. This is because vector $\mathbf{b}$ has only one nonzero component.

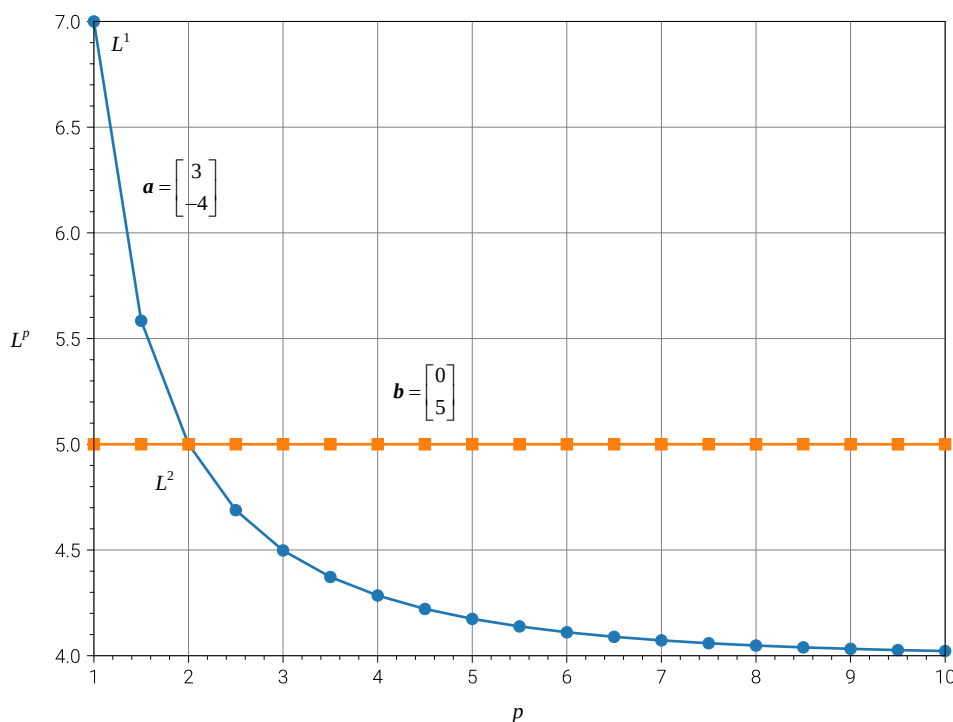


Figure 5. Vectors $\mathbf{a}$ and $\mathbf{b}$'s L^p Norm as a Function of p

Overlaying the L^p Norm Contours

When $p = 1, 2, \infty$, we get the L^1 , L^2 , and L^∞ norms, respectively. We plot the contours where each of these three norms equals 1 on the same figure, as shown in Figure 6.

As shown in Figure 6, the column vector $[1/2, 1/2]^T$ has an L^1 norm of 1, that is,

$$\left\| \begin{bmatrix} 1/2 \\ 1/2 \end{bmatrix} \right\|_{L^1} = |1/2| + |1/2| = 1$$

$\begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}^T$'s L^2 norm is 1, that is,

$$\left\| \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix} \right\|_{L^2} = \sqrt{(\sqrt{2}/2)^2 + (\sqrt{2}/2)^2} = 1$$

$[1, 1]^T$'s L^∞ norm is 1, that is,

$$\left\| \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\|_{L^\infty} = \max(|1|, |1|) = 1$$

Clearly, in Figure 6 the three vectors all lie on the same line, but their vector lengths (magnitude, L^2 norm, Euclidean norm, modulus) are clearly not the same.

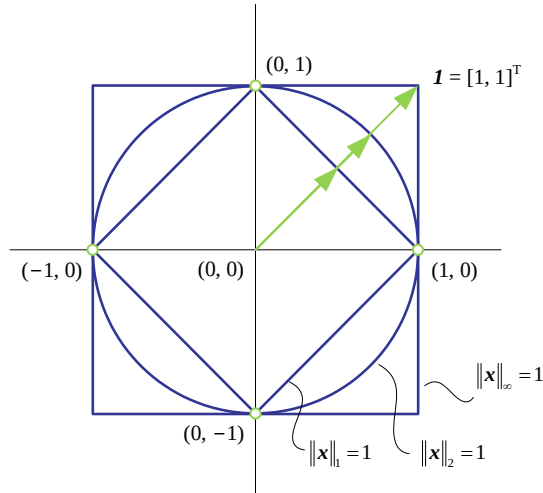


Figure 6. Shapes of the L^1 , L^2 , and L^∞ Norms in the Plane

Also note that in Figure 6, at the four points $(1, 0)$, $(0, 1)$, $(-1, 0)$, and $(0, -1)$, the norm $\|x\|_p$ is always 1.

Except at these four points, as p increases at a given point, the norm $\|x\|_p$ gradually decreases. For example, for the all-ones column vector $[1, 1]^T$, the L^1 norm is 2, the L^2 norm is $\sqrt{2}$, and the L^∞ (infinity) norm is 1. This is something we'll explore further later in this section.

To compare norm sizes more intuitively, we use two perspectives: the same norm value, and the same vector length.

Same Norm Value

To compare different norms more intuitively, we plotted Figure 7. The six subplots each hold the L^p norm value constant (set to 1), and observe how p affects the vectors that satisfy this condition.

As shown in Figure 7 (a), a set of vectors satisfying an L^1 norm of 1; the endpoints of this set of vectors lie on a rotated square. That is, all vectors in Figure 7 (a) satisfy $\|x\|_1 = 1$. The lengths (magnitude, L^2 norm, Euclidean norm, modulus) of these vectors are clearly “uneven.” Only the vectors corresponding to the four points $(1, 0)$, $(0, 1)$, $(-1, 0)$, and $(0, -1)$ have a length (L^2 norm) of 1; other vectors, such as $[1/2, 1/2]^T$, all have a length less than 1.

As shown in Figure 7 (b), the $L^{1.5}$ -norm contour transitions from a rotated square to a circle, with the corners becoming softer; Figure 7 (b) shows a set of vectors satisfying $\|x\|_{1.5} = 1$.



Please compare on your own Figure 7 (b)'s vectors and their length (L^2 norm) characteristics.

As shown in Figure 7 (c), the L^2 -norm contour becomes a circle — this is also the iso-distance line of Euclidean distance; the set of vectors satisfying $\|x\|_2 = 1$ have their endpoints on the unit circle. Clearly, the lengths (magnitude, L^2 norm, Euclidean norm, modulus) of this set of vectors are all exactly the same, equal to 1. The L^2 norm of any vector reflects our intuitive notion of “distance” and “length” — it fundamentally measures the Euclidean distance from a vector to the origin.

As shown in Figure 7 (d) and (e), as p increases to 3 and 5, the shape of the norm's contour becomes increasingly square, with steeper edges, gradually developing toward a more extreme shape. As shown in Figure 7 (f), the L^∞ norm's contour forms a square.

Looking at vector length (magnitude, L^2 norm, Euclidean norm, modulus), the vector lengths (L^2 norm) at the four points $(1, 0)$, $(0, 1)$, $(-1, 0)$, and $(0, -1)$ are still 1; the vector lengths (L^2 norm) of all other vectors are greater than 1.

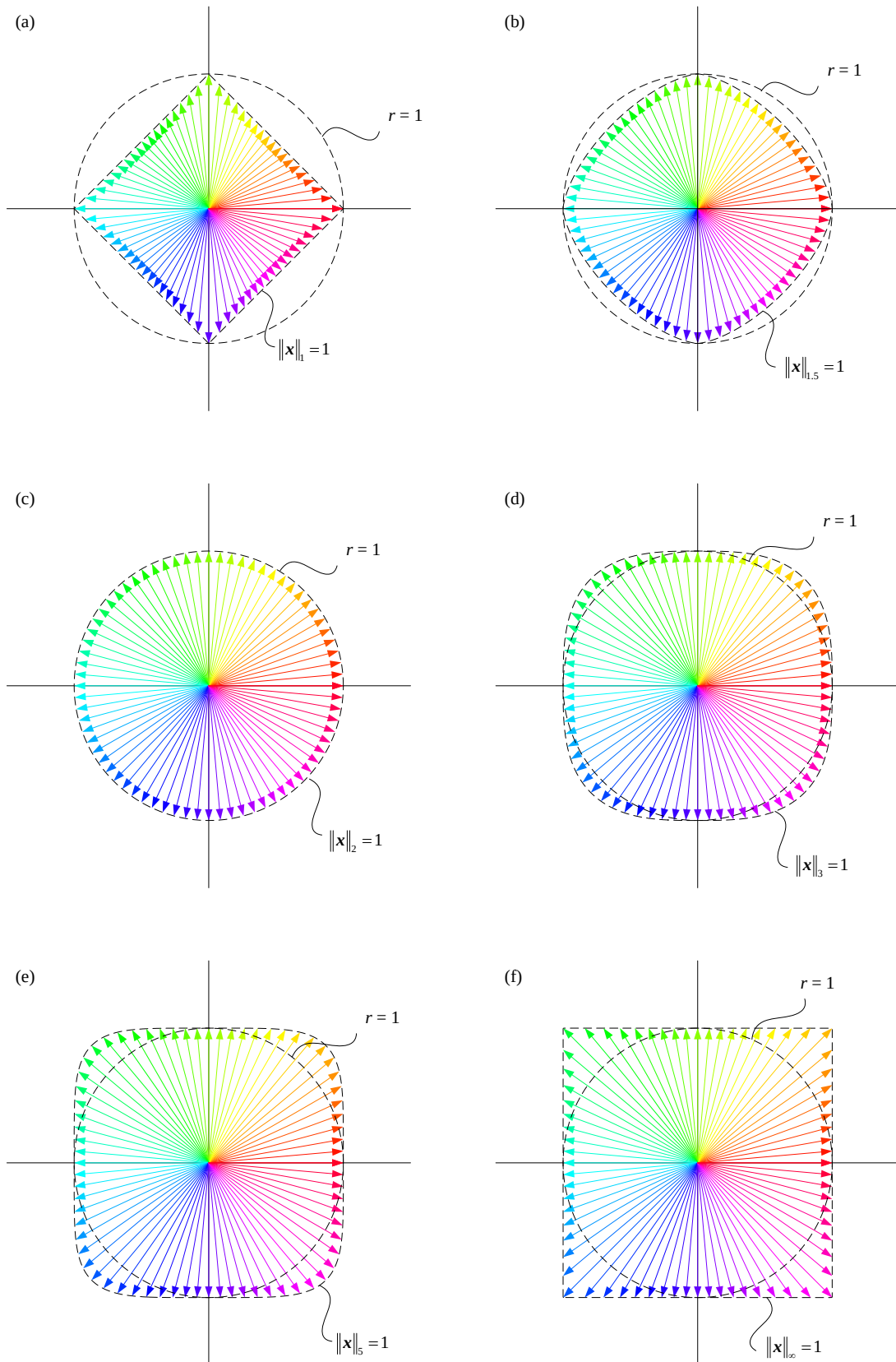


Figure 7. A Set of Vectors with Norm Equal to 1

Same Vector Length

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Now let's compare different vector norms from another angle — the L^p norm size of unit vectors pointing in different directions.

As shown in Figure 8 (a), given a set of unit vectors in the plane; clearly, the vector lengths (magnitude, L^2 norm, Euclidean norm, modulus) of these vectors are all 1. This shows that they have the same length under the Euclidean metric.

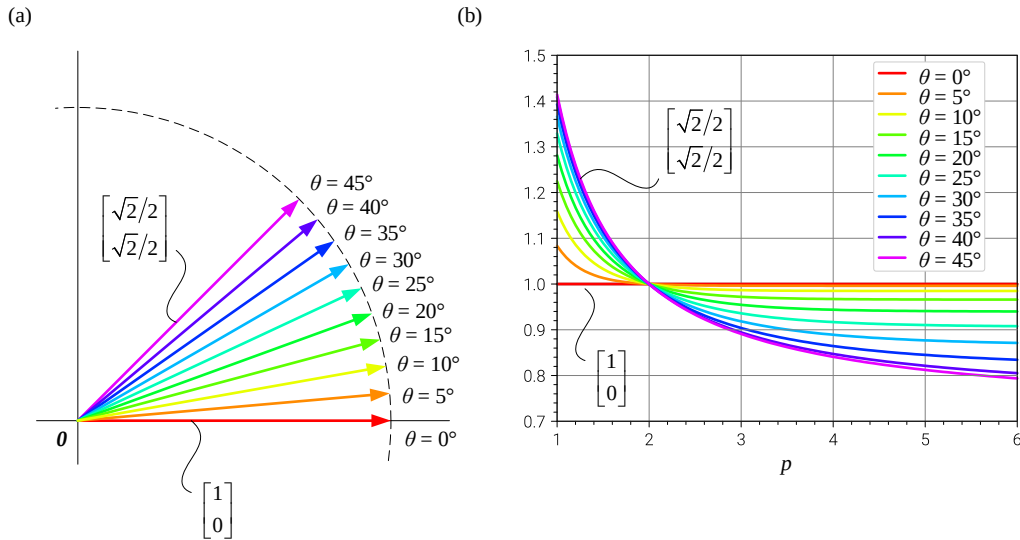


Figure 8. How the L^p Norm of Unit Vectors Changes with p

In addition, this set of vectors lies in the first quadrant, with the angle to the positive horizontal axis gradually changing from 0° to 45° .

However, when we switch to measuring with the L^p norm, the relative “size” of the vectors changes as p changes. Note that the quoted “size” here refers to the L^p norm.

This is visually illustrated in Figure 8 (b), which shows how the norm of this set of unit vectors changes with p .

Figure 8 (b)'s red line tells us that the unit vector $[1, 0]^T$'s L^p norm does not change with p , since its angle with the positive horizontal axis is 0° . $[1, 0]^T$'s L^p norm is

$$\left\| \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\|_p = (1^p + 0^p)^{\frac{1}{p}} = 1$$

$\begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}^T$ This unit vector corresponds to Figure 8 (b)'s “steep” purple curve, meaning its L^p norm is affected most strongly by changes in p . The unit vector $\begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix}^T$ makes a 45° angle with the positive horizontal axis. This vector's L^p norm is

$$\left\| \begin{bmatrix} \sqrt{2}/2 \\ \sqrt{2}/2 \end{bmatrix} \right\|_p = \left(\left(\sqrt{2}/2 \right)^p + \left(\sqrt{2}/2 \right)^p \right)^{\frac{1}{p}} = 2^{\left(\frac{1}{p} - \frac{1}{2} \right)}$$

As p tends toward (approaches) 1, the L^p norm tends toward $\sqrt{2}$; as p tends toward $+\infty$, the L^p norm tends toward $\sqrt{2}/2$; in particular, when $p = 2$, the L^p norm is 1.

Through this comparison, we can gain a deeper understanding of how unit vectors in different directions behave under different norms.

$0 < p < 1$

As shown in Figure 9, for the L^p norm's analytic expression, even when $0 < p < 1$ we can still draw a contour line, though it is no longer a norm in that case. The contour's shape becomes more "pointed," forming a star-like pattern. This geometric shape still has value in some optimization problems or data transformations, even though it doesn't satisfy the strict definition of a norm.

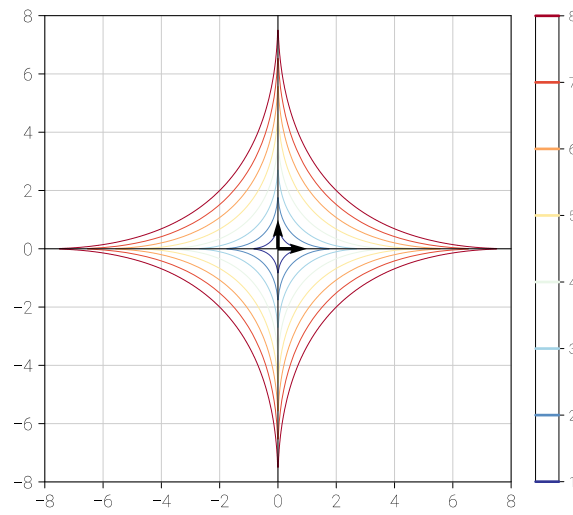


Figure 9. $\left(\sum_{i=1}^n |x_i|^p \right)^{\frac{1}{p}}$ Contour, $p = 0.5$

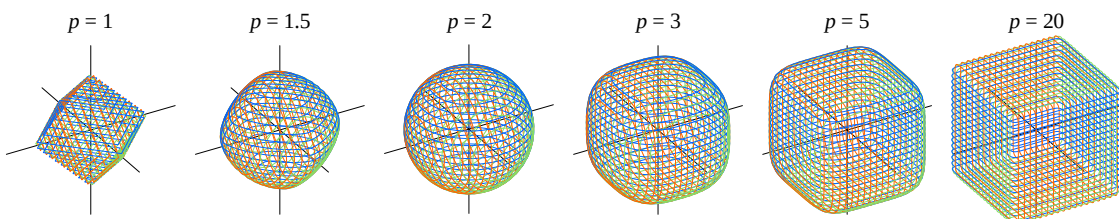
3D Vectors

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

For the 3D vector $\mathbf{x}$, the L^p norm is:

$$\left\| \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right\|_p = \left(|x_1|^p + |x_2|^p + |x_3|^p \right)^{\frac{1}{p}}$$

Figure 10 shows how the iso-distance surface of the L^p norm for a 3D vector changes with p .



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Figure 10. Iso-Distance Surface of the L^p Norm for a 3D Vector as p Varies

For the 3D vector $\mathbf{x} = [x_1, x_2, x_3]^T$, the L^1 norm's iso-distance surface (iso-distance surface) in 3D space is a rotated cube (a special octahedron), satisfying

$$|x_1| + |x_2| + |x_3| = c$$

$\mathbf{x} = [x_1, x_2, x_3]^T$, the L^2 norm's iso-distance surface in 3D space is a sphere, satisfying

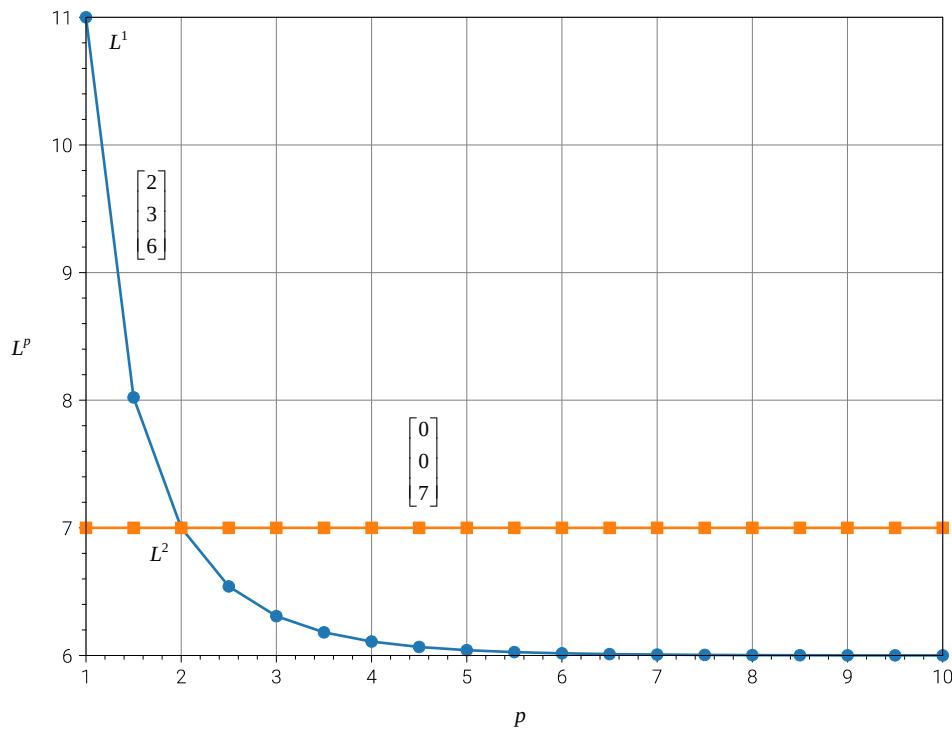
$$x_1^2 + x_2^2 + x_3^2 = c$$

$\mathbf{x} = [x_1, x_2, x_3]^T$, the L^∞ norm's iso-distance surface in 3D space is a cube, satisfying

$$\max(|x_1|, |x_2|, |x_3|) = c$$

Figure 11 shows the vectors $\mathbf{x} = [2, 3, 6]^T$ and $\mathbf{y} = [0, 0, 7]^T$, and how their L^p norms change with p (two curves).

? Please calculate, on your own, the $[2, 3, 6]^T$ and $[0, 0, 7]^T$ vectors' L^1 , L^2 , and L^∞ norms.

Figure 11. Vectors $[2, 3, 6]^T$ and $[0, 0, 7]^T$'s L^p Norm as a Function of p

Vector Norms in RGB Color Space

Let's return to RGB space and look at how the color vectors shown in Figure 12 (a) – their L^p norms – change with p .

Figure 12 (b) shows that the red, green, and blue vectors' L^p norms are all 1, unaffected by p .

Figure 12 (c) shows how the L^p norms of the yellow, magenta, and cyan vectors change with p . As p approaches 1, the L^p norm approaches 2; as p tends toward $+\infty$, the L^p norm approaches 1.

Figure 12 (d) shows how the white vector's L^p norm changes with p . As p approaches 1, the L^p norm approaches 3; as p tends toward $+\infty$, the L^p norm approaches 1.

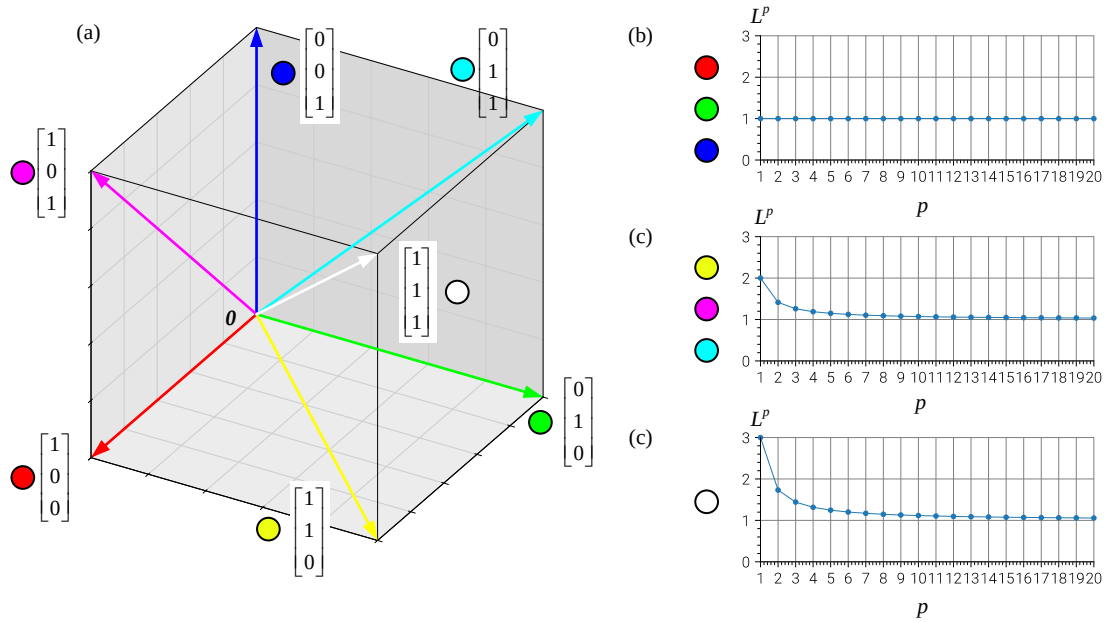


Figure 12. L^p Norms of Common Color Vectors in RGB

Figure 13 shows a set of color vectors in RGB space with an L^1 norm of 1. We can see that the endpoints of this set of vectors lie on a plane. This is easy to understand: these color components satisfy $x_1 + x_2 + x_3 = 1$.

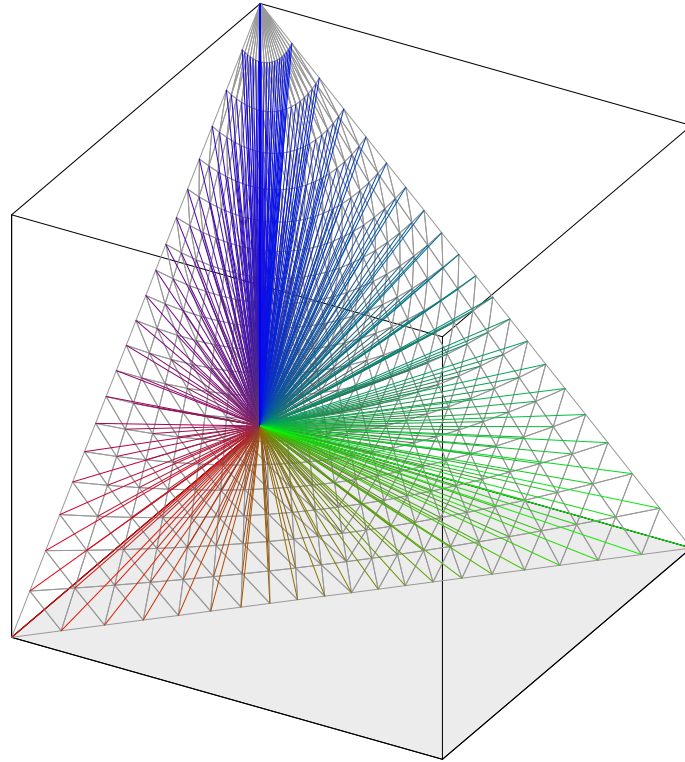
Figure 13. Color Vectors with L^1 Norm Equal to 1 in RGB

Figure 14 shows a set of color vectors in RGB space with an L^2 norm of 1. As you can probably already tell, the endpoints of these vectors lie on a $1/8$ patch of the unit sphere's surface. Figure 15 shows a set of color vectors in RGB space with an L^∞ norm of 1 (including the white vector). These color vectors all lie on the three brightest faces of the RGB color cube.

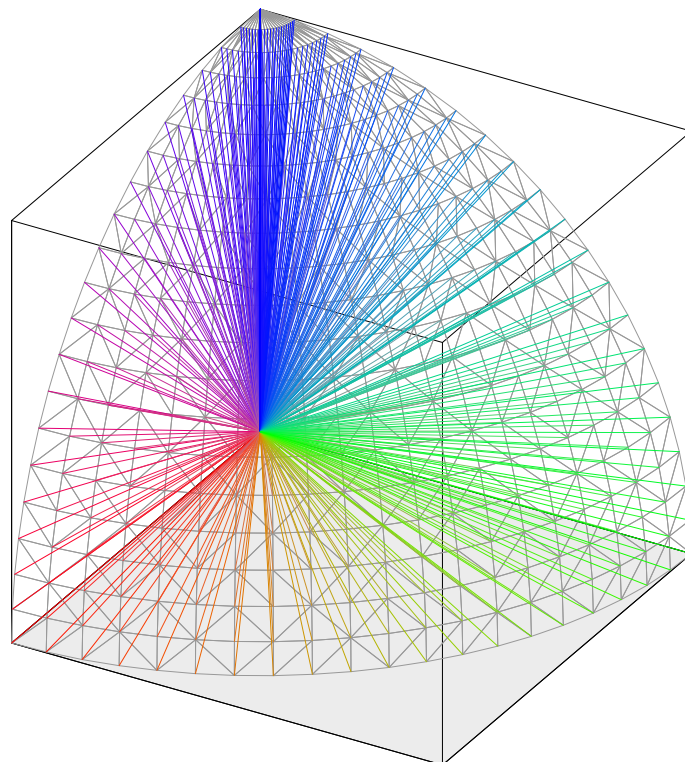
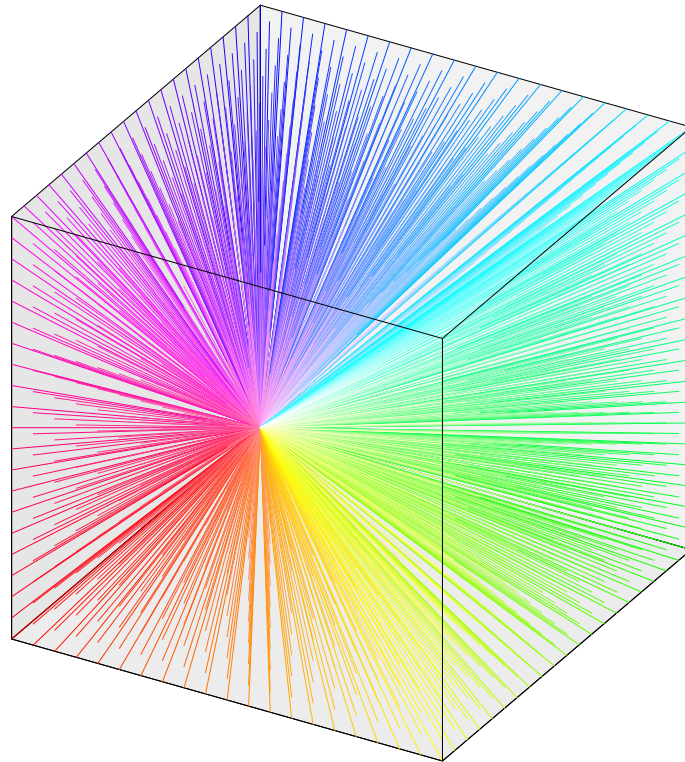


Figure 14. Color Vectors with L^2 Norm Equal to 1 in RGBFigure 15. Color Vectors with L^∞ Norm Equal to 1 in RGB

Distance Metrics

In mathematics, we're often interested in how "far apart" two vectors are.

Given two vectors $\mathbf{a}$ and $\mathbf{b}$ in the plane,

$$\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$

we can measure the difference between them using different "distance definitions." Unless stated otherwise, $\mathbf{a}$ and $\mathbf{b}$ are assumed to start at the origin, and the distance here refers to the distance between the endpoints of $\mathbf{a}$ and $\mathbf{b}$.

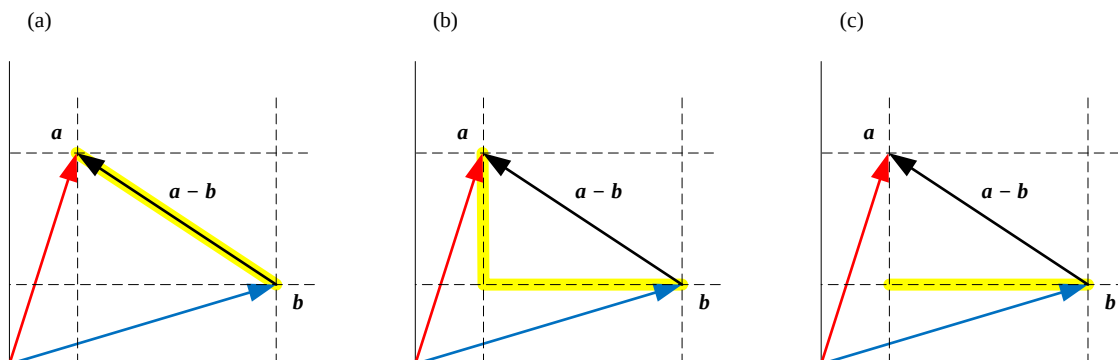


Figure 16. Three Different Distance Metrics

Using the different distance metrics introduced earlier in this section — using the L^2 norm, i.e., the Euclidean norm, to measure the distance between $\mathbf{a}$ and $\mathbf{b}$, we get

$$\|\mathbf{a} - \mathbf{b}\|_2 = \sqrt{(a_1 - b_1)^2 + (a_2 - b_2)^2}$$

As shown in Figure 16 (a), this is the most intuitive notion of distance in everyday life — it's like the straight-line length measured with a ruler.

If we use the L^1 norm to measure the distance between $\mathbf{a}$ and $\mathbf{b}$, we get

$$\|\mathbf{a} - \mathbf{b}\|_1 = \sqrt{|a_1 - b_1| + |a_2 - b_2|}$$

As shown in Figure 16 (b), this is like walking through a grid of city streets, where you can only move horizontally or vertically.

If we use the L^∞ norm to measure the distance between $\mathbf{a}$ and $\mathbf{b}$, we get

$$\|\mathbf{a} - \mathbf{b}\|_\infty = \max(|a_1 - b_1|, |a_2 - b_2|)$$

As shown in Figure 16 (c), the L^∞ norm takes the larger of the two directional offsets as the distance.

There are many more distance metrics used in machine learning — please explore them on your own.

Pairwise Distance Matrix

As shown in Figure 17, we've placed 26 points in the 2D plane, named A, B, C, ..., Z; B and S overlap, and D and O overlap. Clearly, a distance can be computed between every pair of points.

Using combinatorics to count them, there are 325 distance values in total!

How should we record so many distance values?

We can use a matrix!

Figure 18 shows the pairwise distance matrix. The element in row i , column j of this matrix represents the distance between the i -th point and the j -th point.

In the next chapter, we'll formally introduce the concept of the “matrix.” A matrix can not only store distance information like this, but also plays a key role in linear transformations!

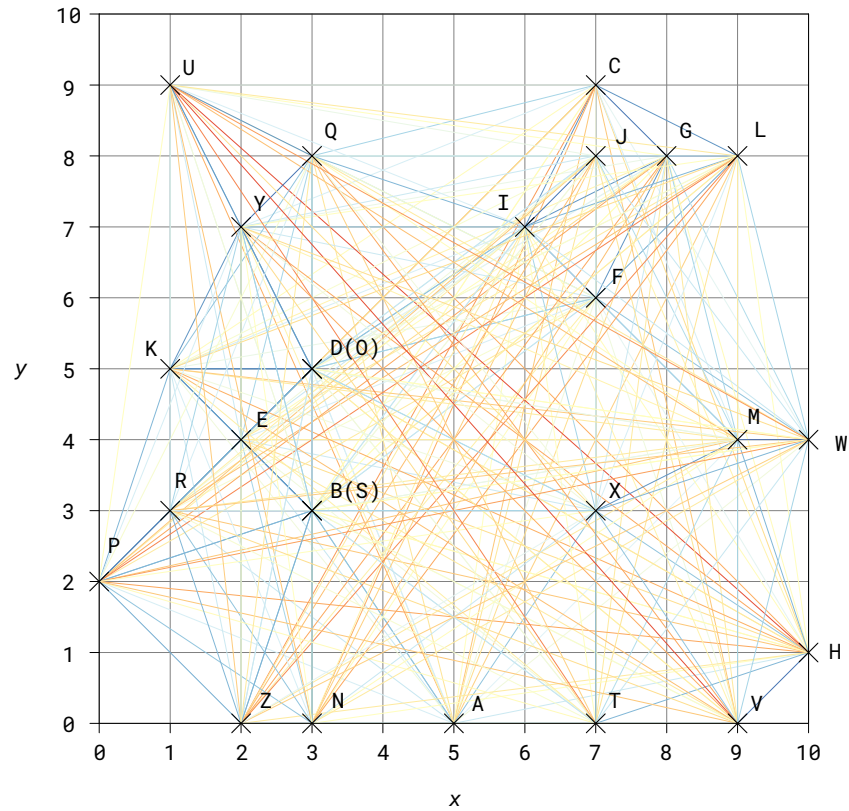


Figure 17. Pairwise Euclidean Distances Among 26 Points in the Plane, Image from Coding Made Easy

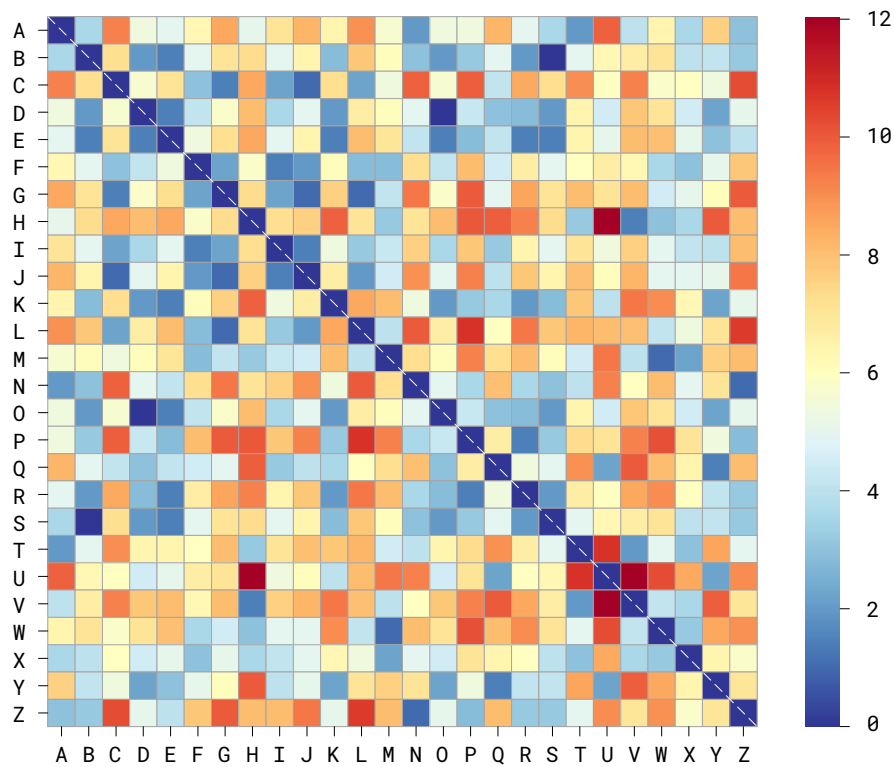


Figure 18. Pairwise Euclidean Distance Matrix, Image from Coding Made Easy



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Learn to use Matplotlib to plot the contour of $f(x_1, x_2) = x_1^2 + x_2^2$.

Q2. Given the vector $[1, 2, 3]^T$, compute by hand its L^1 , L^2 , and L^∞ norms.

Q3. Given the vector $[1, 2, 3]^T$, plot the curve of how the vector's L^p norm changes with p .

Q4. Compare the following pairs of vectors' L^p norms at different values of p , and explain why.

► $[1, 2, 2]^T, [0, 0, 3]^T$

► $[2, 4, 4]^T, [0, 0, 6]^T$

► $[2, 3, 6]^T, [0, 0, 7]^T$

► $[1, 4, 8]^T, [0, 0, 9]^T$

Q5. Please write Python code to plot Figure 5 and Figure 11.

Q6. What is subadditivity, and how does it relate to vector norms?

Q7. Why, when $0 < p < 1$, is $\|\cdot\|_p$ not a norm?

Q8. Please compute the L^1 , L^2 , and L^∞ norms of the following color vectors.

► Red vector $[1, 0, 0]^T$

► Yellow vector $[1, 1, 0]^T$

► White vector $[1, 1, 1]^T$

Q9. Please use `matplotlib.pyplot.contourf()` to try plotting the filled contour plots of Figure 1, Figure 2, and Figure 3.